

Daniel Mouzo Calzadilla

Senior Bioinformatics Scientist & Workflow Architect



SUMMARY

Senior Bioinformatician with 10+ years of combined experience translating complex genomic datasets into scalable, reproducible pipelines. Expert in Nextflow workflow engineering, on-prem HPC cluster management, and leading cross-functional collaborations for €4M international consortia. Fully prepared to design cloud-ready analytical workflows under strict quality and data integrity standards.

CONTACT

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- [linkedin.com/in/daniel.mouzo](https://www.linkedin.com/in/daniel.mouzo)
- github.com/damouzo
- [My Portfolio](#)

CERTIFICATES

- ISO 13485 Internal Auditor (Comply Guru) - 2026
- AWS Cloud Practitioner (AWS) - 2025
- Nextflow Advanced (Sequera) - 2025
- Project Management (Google) - 2023
- Machine learning and Big Data for Bioinformatics (UGR) - 2023
- Cybersecurity (Cámara de Comercio) - 2023
- Data Science Specialized Program (IBM) - 2020

HARD SKILLS

- R
- Linux
- Python
- SQL
- Nextflow
- Git
- ML
- Big Data
- Containers
- AWS
- Project Management

SOFT SKILLS

- Adaptability
- Cross-Functional Collab.
- Problem-solving
- Agile Management
- Scientific Communication

WORK EXPERIENCE

SENIOR BIOINFORMATICIAN

2025-now

Barts Cancer Institute – Queen Mary University

- Engineered production-grade Nextflow pipelines processing 20+ TB of clinical WGS data within the Genomics England.
- Automated reporting systems, integrating variant calling with SQL clinical data to reduce turnaround times to under 3 hours.

POSTDOCTORAL BIOINFORMATICIAN

2021-2025

Navarrabiomed biomedical research center

- Coordinated a cross-functional work package of 30+ multi-site researchers for European gene therapy project.
- Developed automated single-cell NGS pipelines, standardizing and data harmonization across international clinical centers.

SCIENTIFIC COLLABORATION VISITING

2023 & 2025

Ospedale San Raffaele and KAUST Univertisty

- OSR: Harmonized single-cell multiomic clinical datasets to validate safety and cellular integration in advanced gene therapy models.
- KAUST: Designed automated pipelines for NGS preprocessing and deployed relational SQL DBs to centralize experimental metadata.

RESEARCHER ASSIST. & PHD STUDENT

2015-2020

University of Santiago de Compostela

- Engineered quantitative proteomic analytics and LC-MS data pipelines, reducing sample processing by 1.5 days.
- Designed statistical traceability for industrial quality control, delivering significant cost savings for commercial partners.

EDUCATION

PH.D. IN MOLECULAR BIOLOGY ([USC](#))

2017-2020

MD IN INNOV. IN FOOD SAFETY AND TECH. ([USC](#))

2016-2017

DEGREE IN BIOLOGY ([USC](#))

2012-2016

SELECTED PROJECTS

- **XPAND:** Harmonized scMultiome clinical datasets across 3 European hospitals for therapeutic HSC characterization. Patent pending..
- **Genomics England Variant Engine:** Built an automated Nextflow pipeline for clinical variant calling, and high-throughput reporting.
- **Open Source:** Created two R packages adopted by 3 labs for NGS.

LEADERSHIP

BIOINFORMATICS INSTRUCTION: Designed and taught "Data Analysis with R" and "Intro to Omics" courses for 50+ researchers.

SCIENTIFIC COMMUNITY : Founded and coordinated 2 cross-disciplinary biotechnology book clubs.

LANGUAGES

- **SPANISH** Native
- **ENGLISH** Fluent
- **GALICIAN** Native
- **SPANISH SIGN LANGUAGE** Basic